

$$\begin{bmatrix} 1 & y_1/x_1 & z_1/x_1 \\ 0 & 1 & \frac{x_1 z_2 - x_2 z_1}{x_1 x_2 - x_2 x_1} \\ 0 & 1 & \frac{x_1 x_2 - x_2 x_1}{x_1 x_2 - x_2 x_1} \\ 0 & 1 & \frac{x_1 z_4 - x_2 z_1}{x_1 x_4 - x_2 x_1} \end{bmatrix}$$

$$\text{Row 3} = \text{Row 3} - \text{Row 2} \rightarrow$$

$$\begin{bmatrix} x_1 z_2 - x_2 z_1 & x_1 z_2 - x_2 z_1 \\ x_1 x_2 - x_2 x_1 & x_1 x_2 - x_2 x_1 \\ x_1^2 x_2 z_2 - x_1 x_2 x_2 z_1 + x_1 x_2 x_1 z_2 - x_2 x_2 x_1 z_1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & y_1/x_1 & z_1/x_1 \\ 0 & 1 & \frac{x_1 z_2 - x_2 z_1}{x_1 x_2 - x_2 x_1} \\ 0 & 0 & \frac{x_1 z_2 - x_2 z_1}{x_1 x_2 - x_2 x_1} \\ 0 & 1 & \frac{x_1 z_4 - x_2 z_1}{x_1 x_4 - x_2 x_1} \end{bmatrix}$$

$$\text{Row 4} = \text{Row 4} - \text{Row 2} \rightarrow$$

$$\begin{bmatrix} 1 & y_1/x_1 & z_1/x_1 \\ 0 & 1 & \frac{x_1 z_2 - x_2 z_1}{x_1 x_2 - x_2 x_1} \\ 0 & 0 & \frac{x_1 z_2 - x_2 z_1}{x_1 x_2 - x_2 x_1} \\ 0 & 0 & \frac{x_1 z_4 - x_2 z_1}{x_1 x_4 - x_2 x_1} \end{bmatrix}$$

$$\text{Row 3} = \text{Row 3} \times 1/ \left(\frac{x_1 z_2 - x_2 z_1}{x_1 x_2 - x_2 x_1} - \frac{x_1 z_2 - x_2 z_1}{x_1 x_2 - x_2 x_1} \right)$$

$$\begin{bmatrix} 1 & y_1/x_1 & z_1/x_1 \\ 0 & 1 & \frac{x_1 z_2 - x_2 z_1}{x_1 x_2 - x_2 x_1} \\ 0 & 0 & 1 \\ 0 & 0 & \frac{x_1 z_4 - x_2 z_1}{x_1 x_4 - x_2 x_1} \end{bmatrix}$$

$$\text{Row 4} = \text{Row 4} \times 1/ \left(\frac{x_1 z_4 - x_2 z_1}{x_1 x_4 - x_2 x_1} - \frac{x_1 z_2 - x_2 z_1}{x_1 x_2 - x_2 x_1} \right)$$

$$\begin{bmatrix} 1 & y_1/x_1 & z_1/x_1 \\ 0 & 1 & \frac{x_1 z_2 - x_2 z_1}{x_1 x_2 - x_2 x_1} \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Row 4} = \text{Row 4} - \text{Row 3}$$

$$\begin{bmatrix} 1 & y_1/x_1 & z_1/x_1 \\ 0 & 1 & \frac{x_1 z_2 - x_2 z_1}{x_1 x_2 - x_2 x_1} \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$